JUAN FELIPE MALAVER

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Analytics Consultant & Data Scientist specialized in advanced quantitative modeling, applied Machine Learning, and strategic consulting. Trained at the London School of Economics and Universidad del Rosario, with a Master's in Business Analytics (fully funded by scholarship). I combine a solid foundation in econometrics and statistics with technical skills in Python, R, SQL, AWS, and the deployment of analytical solutions. Experienced leading high-impact projects at Mastercard, translating complex algorithms into profitable commercial strategies for senior banking leadership.

PROFESSIONAL EXPERIENCE

Mastercard
Bogotá, Colombia

Associate Consultant | Strategy & Analytics
December 2025 - Present

- Developed robust ETL pipelines in Python and SQL to automate the ingestion of large-scale transactional data, optimizing the team's analytical performance working with PII always under GDPR norms.
- Designed and implemented Machine Learning solutions (K-Medoids Clustering with Gower Distance) and optimization models to segment merchants, generating over a projected COP 5,000 million (1.6M USD) in annual profitability following commercial deployment.
- Applied causal inference models (Propensity Score Matching and Difference-in-Differences) in Python to evaluate personalization campaigns, increasing the average spend lift by 3% per iteration.
- Translated predictive models and data science metrics into executive visualizations for bank boards of directors, facilitating data-driven strategic decisions.

Associate Consultant Intern
January 2025 - December 2025

- Built commercial projection and predictive models in Python, R, and SQL to optimize mass campaigns in Latin American banking, facilitating data ingestion, feature engineering, data governance and data quality protocols.
- Designed segmentation and scoring methodologies based on RFM models, increasing the monetary effectiveness of loyalty programs by 23%.
- Built an ensemble model using classification algorithms (Random Forest and XGBoost) to optimize communication strategy and flag propensity of clients to buy certain products depending on their characteristics, translating the model in archetypes for the marketing team.

Universidad del Rosario
Bogotá, Colombia

Econometric Research Assistant
January 2024 - January 2025

- As an outsourced contractor, developed automations and advanced econometric models (Callaway-Sant'Anna DiD) for research projects with Princeton, Penn State, and the University of Chicago; processed geospatial databases and used Python/R for satellite image analysis.

Teaching Assistant in Statistics
January 2023 - January 2024

- Taught university courses with four weekly hours of lectures and two hours of office hours for Economics and Applied Mathematics to Computer Science students. The course was based on R and Python coding and covered sampling designs, bias, descriptive statistics, data visualization, probability, statistical distributions, inference, hypothesis testing, regression and modeling, panel data, DiD, PSM, and Monte Carlo.

EDUCATION

London School of Economics (LSE)
London, United Kingdom

Undergraduate Degree in Economics
January 2021 - October 2025

- Upper Second-Class Honours | Minor in Poverty and Inequality | Student Voice Group Champion, 2022-2023

Universidad del Rosario
Bogotá, Colombia

Undergraduate Degree in Economics
July 2020 - October 2025

- Minor in International Trade and Finance | GPA: 4.6/5.0 | Graduated with academic honors | Selected to deliver the graduation speech | Academic scholarship
- Best final project in Applied Microeconomics in the faculty, 2021

Masters in business Analytics
January 2026 - In progress

- 100% academic scholarship | GPA: 4.8/5.0

PROJECTS AND INDEPENDENT PORTFOLIO

TreeStrat - Decision Tree Ensemble Business Studio (Streamlit)

- Developed an interactive Streamlit application for decision tree ensemble modeling aimed at business forecasting and scenario simulation, translating complex variable interactions into actionable business rules and revenue optimization.

ClusterLab - Adaptive Clustering Tool Business (Vercel)

- Built and deployed an interactive web tool for dynamic customer segmentation using unsupervised clustering algorithms, enabling marketing and product teams to visualize clusters in real time and maximize the ROI of commercial campaigns, specifically optimized for non-technical users and SME owners.

ML_for_ALL – Educational Repository in R for non-technical users that want to learn ML

- A GitHub repository featuring synthetic datasets designed to provide non-technical users with an introduction to machine learning, focusing on data balancing, logistic regression, classification algorithms, ensemble models for upselling, and time series analysis. It includes detailed explanations throughout the code and tips for understanding the potential business applications of each algorithm.

AWARDS AND RECOGNITIONS

For more details on my other certifications please refer to LinkedIn

Award to National Excellence | *Ministry of Education | November 2025*

- Recognized for outstanding performance on the national Saber Pro examination, ranking 35th nationwide across all degree programs and 5th among Economics students

Award of Excellence | *Universidad del Rosario | November 2025*

- Recognized among the university's most outstanding alumni for excellence and commitment across academic, cultural, athletic, social, research, and entrepreneurial activities

Full Scholarship for Graduate Studies | *Universidad del Rosario | July 2025*

- Awarded a 100% tuition scholarship for graduate studies in recognition of academic merit

Global Innovation Challenge Finalist | *Mastercard | July 2025*

- Selected as a global finalist in Mastercard's innovation competition. Led the design of Innova, an AI-powered platform leveraging Mastercard data to increase card payment adoption among SMEs through merchant recommendations, budgeting tools, customer segmentation, and personalized insight

Onboarding Case Winners | *Mastercard | April 2025*

- Competed against teams across Latin America and the Caribbean in a regional consulting case competition, contributing as data analyst and presenter

Best intern of the year | *Mastercard - Universidad del Rosario | 2025*

- Recognized as Intern of the Year through a joint university-industry award based on professional performance and institutional evaluation

Semifinalist in NCC - National Consulting Case | *National | November 2024*

- Represented Universidad del Rosario in a national consulting competition, advancing to the semifinal stage and tackling business cases proposed by BCG while providing advisory support to companies including Pavco

4th Place in the National Academic Economics Competition, CANE | *National | 2024*

- Selected to represent Universidad del Rosario based on top faculty GPA in econometrics and statistics; placed 4th nationally in the economics competition

LANGUAGES – STACK AND SKILLS

Spanish: Native | **English**: C1 (IELTS Band 7.5, certified) | **French**: Basic knowledge (A1/A2) | **Italian**: Basic knowledge (A1/A2)

- **Python (Advanced):** ETL pipeline development, predictive modeling, automation, and machine learning using Scikit-Learn, Pandas, NumPy, PyTorch, PySpark, Jupyter, TensorFlow, and VS Code.
- **R (Advanced):** Advanced econometric modeling, time series, spatial analysis, and big data processing.
- **SQL (Advanced):** Complex queries for data extraction and manipulation at large scale (Hadoop, Impala, SQLite, MySQL).
- **AWS (Basic / Preparing for MLA-C01 certification):** Data preparation, design, and deployment of Machine Learning workflows and Amazon SageMaker, S3, Athena, DataWrangler, AmazonGlue.
- **Data ingestion Platforms (Databricks, Dataiku, Knime):** Data analysis, Business Intelligence, and enterprise ETL workflows.
- **Predictive Modeling and Classification:** Supervised and unsupervised algorithms, segmentation, and advanced clustering (K-Medoids, Gower Distance).
- **Causal Inference and Econometrics:** Propensity Score Matching (PSM), Difference-in-Differences (DiD, Callaway-Sant'Anna approach), and Monte Carlo simulations.
- **Streamlit and Vercel (Intermediate/Basic):** Rapid creation and deployment of interactive applications and business-oriented analytical tools (e.g., TreeStrat Business Studio and Adaptive Clustering Tool).
- **Business Intelligence (Tableau, Power BI):** Design of executive dashboards and interactive boards for C-level decision-making.
- **Office Tools:** Advanced Excel (Macros, Solver, Power Query, Power Pivot), LaTeX for technical and academic documentation.
- **AI:** MCP, Skills, Agents – agent/sub-agent teams, development via Antigravity, prompt engineering, instruction customization, vector databases, CrewAI, embeddings and fine-tuning, AI governance and risk.

MAIN CERTIFICATIONS

For more details on my other certifications please refer to LinkedIn

Leveraging Machine Learning for Business — *Wharton - 2026*

Lean Six Sigma Yellow Belt - LSS — *Universidad de los Andes - Ongoing*

MLA -C01 (Machine Learning Associate Engineer) — *AWS - Ongoing*